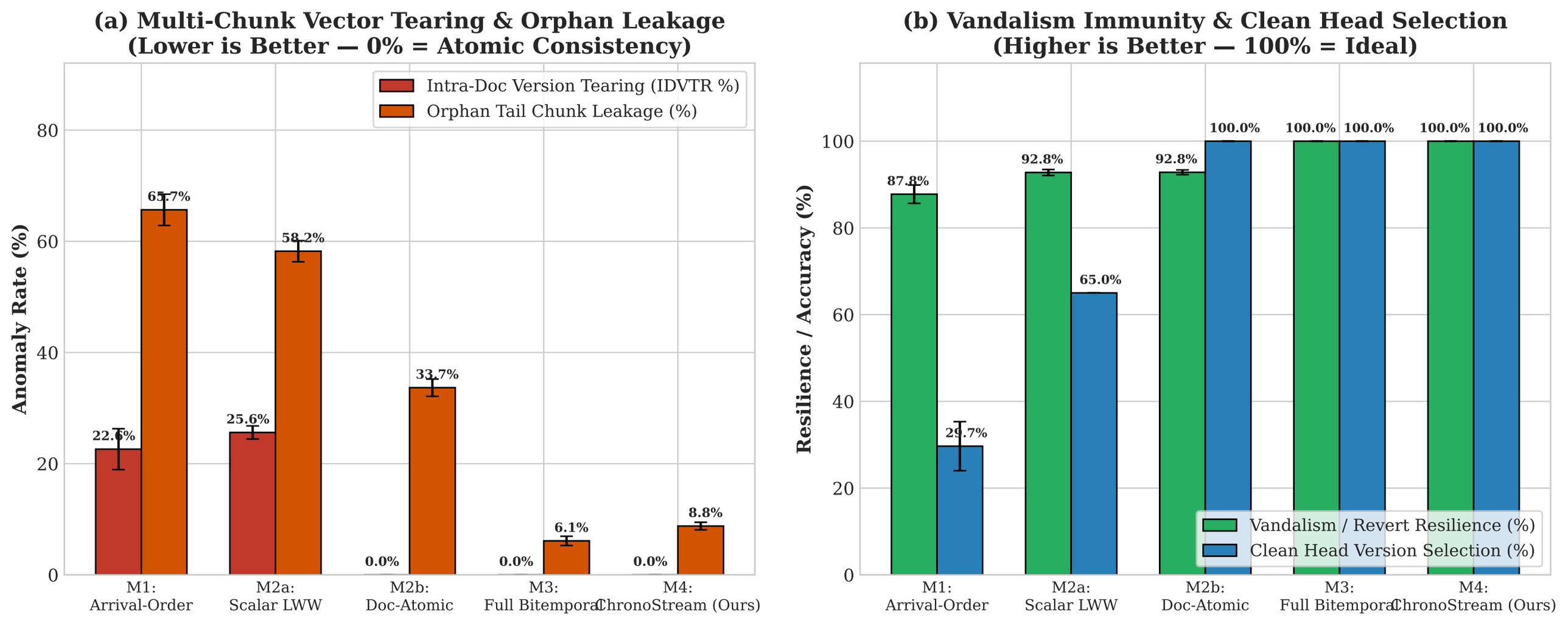
(a) Multi-Chunk Vector Tearing & Orphan Leakage
(Lower is Better — 0% = Atomic Consistency)
Anomaly Rate (%)
Intra-Doc Version Tearing (IDVTR %)
Orphan Tail Chunk Leakage (%)
22.6%
65.7%
25.6%
58.2%
0.0%
33.7%
0.0%
6.1%
0.0%
8.8%
M1: Arrival-Order
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitempora
M4: ChronoStream (Ours)
(b) Vandalism Immunity & Clean Head Selection
(Higher is Better — 100% = Ideal)
Resilience / Accuracy (%)
87.8%
29.7%
92.8%
65.0%
92.8%
100.0%
100.0%
100.0%
100.0%
100.0%
Vandalism / Revert Resilience (%)
Clean Head Version Selection (%)
M1: Arrival-Order
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitempora
M4: ChronoStream (Ours)